SBT-DF204 — Computer Forensics Case Study

Case Study 2 Reconstructing Chrome Web History

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Lab Title: Reconstructing Chrome Web History Case Study Guide
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2. Executive conclusion

The examination of the Chrome History database identified 74 recorded browsing visits on 19 April 2022, between approximately 13:54:18 UTC and 15:02:56 UTC. The records included activity involving Craigslist, Gmail, an Imgur-related download, blockchain-explorer pages, and Kraken-related pages.

The analysis was conducted using a working copy of the History database, whose SHA-256 hash matched that of the original evidence file. Four SQL queries were used to examine the chronological browsing timeline, navigation transition values, recorded navigation relationships, and download URL chain. One download record referenced a file named `proof_of_payment.png`, with a recorded size of 33,844 bytes and an associated Imgur image URL.

These findings establish that the examined browser database contains records of the listed activity. However, browser-history evidence alone cannot establish who performed the actions, whether a particular account was successfully accessed, whether an email or online post was submitted, or whether a cryptocurrency transaction was completed. The actual contents of the downloaded image were also not established through the History database analysis alone.

Overall, the examination provides moderate confidence in the identified browser activity but low confidence in attributing that activity to a specific individual without independent corroboration. Further examination of the downloaded file, relevant email artifacts, operating-system records, and reliable account or transaction records is recommended before drawing stronger conclusions about identity, intent, or financial activity.

3. Evidence acquisition and integrity

Item	Record
Source of History file	History. 10/12/2026
Original file name	History
File size (bytes)	196608
SHA-256 (original)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
SHA-256 (working copy)	b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19
Location of read-only original	/home/kali/SBT-DF204/Case2/original/History — permissions set using chmod 444
Location of working copy	/home/kali/SBT-DF204/Case2/working/History_working
Working-copy method	Created using the Linux cp command; SHA-256 verified against the original
File-access precautions	Original file made read-only; working copy opened using SQLite's -readonly option; database tables and records queried without write operations; URLs were not visited
Tools and versions used	Kali Linux; SQLite 3.53.4; sha256sum for SHA-256 hashing; GNU stat for file-size verification

4. Database Method

4.1 Tables and Fields Reviewed

Table	Role in this case	Key field used
urls	Stores website addresses and summary information about browsing activity.	id, url, title, visit_count, last_visit_time
visits	Stores individual browsing events and links between visits.	id, url, visit_time, from_visit, transition
download	Stores information about downloaded files.	id, target_path, current_path, start_time, end_time,

		received_bytes, total_bytes, state, tab_url, referrer
downloads_url_chains	Stores the URL sequence associated with each download.	id, chain_index, url
keyword_search_terms	Stores search terms associated with browser URL records.	keyword_id, url_id, term, normalized_term

The visits.url field contains a numeric identifier that links to urls.id, allowing individual visits to be associated with website addresses. The visits.from_visit field refers to another visit record and can help establish navigation sequences. The downloads_url_chains.id field links download URL-chain records to downloads.id. The keyword_search_terms.url_id field associates search terms with URL records.

The schema contains timestamp fields including last_visit_time in urls, visit_time in visits, and start_time and end_time in downloads. These fields can support the timeline analysis in Section 4.2.

The record counts were 57 rows in urls, 74 in visits, one in downloads, and one in downloads_url_chains. This indicates that the database contains more individual visit records than distinct URL records and includes one download record with one associated URL-chain record. All queries were performed against the working copy in read-only mode.

4.2 Timestamp conversion

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT datetime(11644473600*1000000/1000000 - 11644473600, 'unixepoch') AS epoch_test;"
    epoch_test
-------------------
1970-01-01 00:00:00

kali@kali:~/SBT-DF204/Case2$
```

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT u.id,
        u.last_visit_time,
        MAX(v.visit_time) AS max_visit,
        CASE WHEN u.last_visit_time = MAX(v.visit_time)
             THEN 'MATCH'
             ELSE 'CHECK'
        END AS verification
 FROM urls u
 JOIN visits v ON v.url = u.id
 GROUP BY u.id
 ORDER BY u.id
 LIMIT 10;"
id    last_visit_time       max_visit       verification
--  ------------------  ------------------  ------------
 1   13294850059218619   13294850059218619   MATCH
 2   13294850103307397   13294850103307397   MATCH
 3   13294850059946032   13294850059946032   MATCH
 4   13294850103307397   13294850103307397   MATCH
 5   13294850079868149   13294850079868149   MATCH
 6   13294850062411492   13294850062411492   MATCH
 7   13294850105971574   13294850105971574   MATCH
 8   13294850302757437   13294850302757437   MATCH
 9   13294850105971574   13294850105971574   MATCH
10   13294850110244655   13294850110244655   MATCH

kali@kali:~/SBT-DF204/Case2$
```

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT id,
        target_path,
        start_time,
        datetime(start_time/1000000 - 11644473600, 'unixepoch') AS start_utc,
        end_time,
        CASE WHEN end_time = 0 THEN 'No completion time recorded'
             ELSE datetime(end_time/1000000 - 11644473600, 'unixepoch')
        END AS end_utc,
        received_bytes,
        total_bytes,
        state
 FROM downloads;"
id target_path  start_tim start_utc      end_time      end_utc   received_bytes total_bytes state
                   e
-- ------------ --------- ---------- ----------------- ---------- -------------- ------------ -----
 1 C:\Users\    132948538 2022-04-19 1329485383026132 2022-04-19          33844        33844     1
   FSCS_User\    18494344 14:56:58                  2 14:57:10
   Desktop\
   proof_of_pay
   ment.png
```

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT id,
        datetime(start_time/1000000 - 11644473600, 'unixepoch') AS start_utc,
        datetime(end_time/1000000 - 11644473600, 'unixepoch') AS end_utc,
        received_bytes,
        total_bytes,
        state,
        target_path
 FROM downloads;"
id      start_utc           end_utc       received_bytes total_bytes state      target_path
-- ------------------- ------------------- -------------- ------------ -----  ------------------
 1 2022-04-19 14:56:58 2022-04-19 14:57:10          33844        33844     1 C:\Users\FSCS_User\
                                                                              Desktop\proof_of_paym
                                                                              ent.png
```

Timestamp Validation and Download Analysis

The Chrome History database timestamp conversion was validated using a known epoch value. The test returned 1970-01-01 00:00:00, confirming that the conversion formula produced the expected result. A cross-check of ten URL records also showed that each displayed last_visit_time matched the maximum corresponding visit_time.

The downloads table contained one record for proof_of_payment.png. The database recorded a start time of 19 April 2022 at 14:56:58 UTC and an end time of 14:57:10 UTC. The recorded received and total sizes were both 33,844 bytes. These findings describe the values stored in the browser database and should not, by themselves, be treated as proof of the file's contents or the user's actions.

4.3 Joins and queries

Reg	Purpose / SQL summary	Tables joined	Output ref
Q1	List visits in chronological order with URL, title and UTC time	visits, urls	Figure A8
Q2	Separate navigation transition base type and qualifier bits	visits, urls	Figure A9
Q3	Trace recorded navigation relationships using from_visit	visits (self-join), urls	Figure A10
Q4	Link download details to the associated URL chain	downloads, downloads_url_chains	Figure A11

Four SQL queries were executed against the read-only working copy of the Chrome History database. Q1 joined the visits and urls tables to produce a chronological browsing timeline in UTC. Q2 examined navigation transition values by separating the base type from the qualifier bits. Q3 used left joins to associate visits with their recorded predecessor visits through the from_visit field. Q4 joined the downloads and downloads_url_chains tables to examine the recorded download destination and associated URL chain.

The queries identified browser-history records relating to Craigslist, Gmail, blockchain-explorer searches, Kraken pages, and a download named proof_of_payment.png. These results describe activity recorded in

the browser database. They do not, on their own, establish the identity of the person using the browser, the contents of the downloaded file, or whether a cryptocurrency transaction was completed.

4.4 Transition interpretation

The Chrome visits table contains a transition value that records how a navigation was initiated. I derived the base type using the SQL bitwise operation transition & 255 (equivalent to transition & 0xFF). I derived the qualifier bits using transition & 4294967040 (equivalent to transition & 0xFFFFFF00). Because some values appear negative in SQLite, I interpreted those qualifier results using their unsigned 32-bit representation where necessary.

I also examined the from_visit field to identify the previous visit recorded by Chrome. I cross-checked the transition values against the associated URLs, page titles, timestamps, and recorded navigation relationships. I did not rely on the transition value alone to determine the user's intention.

Raw value	Base type	Qualifiers observed	Interpretation used
805306373	5	805306368 (0x30000000)	Generated navigation. Checked the associated URL, title, and predecessor visit before interpreting the context.
805306368	0	805306368 (0x30000000)	Link-type base value. The qualifier bits were recorded separately and were not treated as proof of a specific user action.
268435456	0	268435456 (0x10000000)	Link-type base value with additional qualifier bits. Cross-checked with the corresponding URL and navigation relationship.
805306375	7	805306368 (0x30000000)	Form-submit base type. Compared with the URL, title, and from_visit relationship to assess the navigation context.
-2147483648	0	2147483648 (0x80000000, unsigned)	Base type is zero after masking. The high-order qualifier bit was considered separately; the signed raw value was not interpreted as a standalone action.
-1610612736	0	2684354560 (0xA0000000, unsigned)	Base type is zero after masking. The qualifier bits were retained in unsigned form and interpreted cautiously alongside the related records.

How from_visit was used: I joined visits.from_visit to visits.id, then joined the predecessor visit to urls to retrieve its URL. This helped show the navigation relationship recorded in the database. A zero or unmatched predecessor did not provide a prior URL through this join.

Interpretation limitation: Chrome transition values describe recorded navigation types and qualifiers; they do not independently prove who performed an action or why. Likewise, from_visit represents a recorded browser navigation relationship, which is not necessarily the same as an HTTP referrer. All conclusions were cross-checked against the relevant URLs, titles, and timestamps.

5. Findings

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF204/Case2/{original,working,screenshots,reports}

┌──(kali㉿kali)-[~]
└─$ cd ~/SBT-DF204/Case2
pwd
find . -maxdepth 2 -type d
/home/kali/SBT-DF204/Case2
.
./working
./screenshots
./reports
./original
```

```
┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ cp /home/kali/Downloads/History ~/SBT-DF204/Case2/original/

┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ ls -lh ~/SBT-DF204/Case2/original/
total 192K
-rw-rw-r-- 1 kali kali 192K Oct 12 10:04 History

┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ file ~/SBT-DF204/Case2/original/History
sha256sum /home/kali/Downloads/History ~/SBT-DF204/Case2/original/History
/home/kali/SBT-DF204/Case2/original/History: SQLite 3.x database, last written using SQLite versi
on 3036000, file counter 33, database pages 41, cookie 0×a7, schema 4, UTF-8, version-valid-for 3
3
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  /home/kali/Downloads/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  /home/kali/SBT-DF204/Case2/orig
inal/History

┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ cp ~/SBT-DF204/Case2/original/History ~/SBT-DF204/Case2/working/History_working

┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ sha256sum ~/SBT-DF204/Case2/original/History ~/SBT-DF204/Case2/working/History_working
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  /home/kali/SBT-DF204/Case2/orig
inal/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  /home/kali/SBT-DF204/Case2/work
ing/History_working

┌──(kali㉿kali)-[~/SBT-DF204/Case2/original]
└─$ sqlite3 ~/SBT-DF204/Case2/working/History_working ".tables"
clusters              downloads               keyword_search_terms    typed_url_sync_metadata
clusters_and_visits   downloads_reroute_info  meta                    urls
content_annotations   downloads_slices        segment_usage           visit_source
context_annotations   downloads_url_chains    segments                visits
```

```
kali@kali:~/SBT-DF204/Case2/original$ stat -c '%s' History
196608

kali@kali:~/SBT-DF204/Case2/original$ sha256sum History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  History

kali@kali:~/SBT-DF204/Case2/original$
```

```
kali@kali:~/SBT-DF204/Case2/original$ chmod 444 History
ls -l History
realpath History
-r--r--r-- 1 kali kali 196608 Oct 12 10:04 History
/home/kali/SBT-DF204/Case2/original/History

kali@kali:~/SBT-DF204/Case2/original$ cp History ../working/History_working
sha256sum History ../working/History_working
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  ../working/History_working

kali@kali:~/SBT-DF204/Case2/original$
```

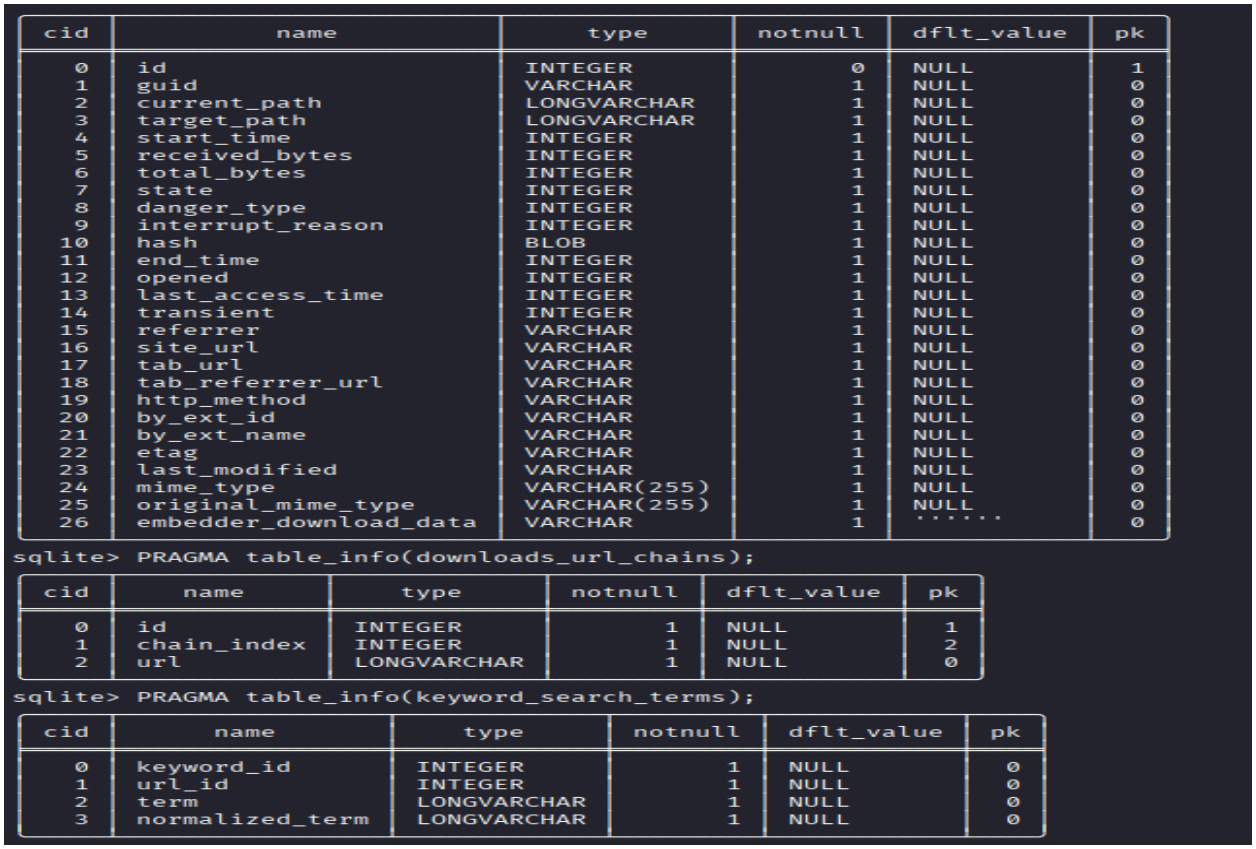
A1

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly \
~/SBT-DF204/Case2/working/History_working
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> PRAGMA table_info(urls);
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	url	LONGVARCHAR	0	NULL	0
2	title	LONGVARCHAR	0	NULL	0
3	visit_count	INTEGER	1	'0'	0
4	typed_count	INTEGER	1	'0'	0
5	last_visit_time	INTEGER	1	NULL	0
6	hidden	INTEGER	1	'0'	0

```
sqlite> PRAGMA table_info(visits);
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	url	INTEGER	1	NULL	0
2	visit_time	INTEGER	1	NULL	0
3	from_visit	INTEGER	0	NULL	0
4	transition	INTEGER	1	'0'	0
5	segment_id	INTEGER	0	NULL	0
6	visit_duration	INTEGER	1	'0'	0
7	incremented_omnibox_typed_score	BOOLEAN	1	FALSE	0
8	opener_visit	INTEGER	0	NULL	0



cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	0	NULL	1
1	guid	VARCHAR	1	NULL	0
2	current_path	LONGVARCHAR	1	NULL	0
3	target_path	LONGVARCHAR	1	NULL	0
4	start_time	INTEGER	1	NULL	0
5	received_bytes	INTEGER	1	NULL	0
6	total_bytes	INTEGER	1	NULL	0
7	state	INTEGER	1	NULL	0
8	danger_type	INTEGER	1	NULL	0
9	interrupt_reason	INTEGER	1	NULL	0
10	hash	BLOB	1	NULL	0
11	end_time	INTEGER	1	NULL	0
12	opened	INTEGER	1	NULL	0
13	last_access_time	INTEGER	1	NULL	0
14	transient	INTEGER	1	NULL	0
15	referrer	VARCHAR	1	NULL	0
16	site_url	VARCHAR	1	NULL	0
17	tab_url	VARCHAR	1	NULL	0
18	tab_referrer_url	VARCHAR	1	NULL	0
19	http_method	VARCHAR	1	NULL	0
20	by_ext_id	VARCHAR	1	NULL	0
21	by_ext_name	VARCHAR	1	NULL	0
22	etag	VARCHAR	1	NULL	0
23	last_modified	VARCHAR	1	NULL	0
24	mime_type	VARCHAR(255)	1	NULL	0
25	original_mime_type	VARCHAR(255)	1	NULL	0
26	embedder_download_data	VARCHAR	1	'.....'	0

```
sqlite> PRAGMA table_info(downloads_url_chains);
```

cid	name	type	notnull	dflt_value	pk
0	id	INTEGER	1	NULL	1
1	chain_index	INTEGER	1	NULL	2
2	url	LONGVARCHAR	1	NULL	0

```
sqlite> PRAGMA table_info(keyword_search_terms);
```

cid	name	type	notnull	dflt_value	pk
0	keyword_id	INTEGER	1	NULL	0
1	url_id	INTEGER	1	NULL	0
2	term	LONGVARCHAR	1	NULL	0
3	normalized_term	LONGVARCHAR	1	NULL	0

The Chrome History database was examined using SQLite version 3.53.4 in read-only mode. The tables were identified using a query on sqlite_master, and their column names and data types were inspected using PRAGMA table_info. The following tables were reviewed to understand the browser activity and download records available for analysis.

Evidence Preservation and Integrity

The original browser History file, with a size of 196,608 bytes, was preserved at /home/kali/SBT-DF204/Case2/original/History. Its SHA-256 hash, recorded before examination, was b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19. The original file was made read-only using the Linux chmod 444 command. A working copy was created using the cp command and stored at /home/kali/SBT-DF204/Case2/working/History_working. The SHA-256 hash of the working copy matched the original hash, confirming that both files had identical contents at the time of verification. Further database examination will be conducted on the working copy in read-only mode.

A2

```
kali@kali:~/SBT-DF204/Case2/original$ sqlite3 -readonly ../working/History_working
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .tables
clusters              downloads                 keyword_search_terms    typed_url_sync_metadata
clusters_and_visits   downloads_reroute_info    meta                    urls
content_annotations   downloads_slices          segment_usage           visit_source
context_annotations   downloads_url_chains      segments                visits
sqlite> .quite
Error: unknown command or invalid arguments:  "quite". Enter ".help" for help
sqlite> .quit

kali@kali:~/SBT-DF204/Case2/original$ sqlite3 --version
uname -a
sha256sum --version | head -n 1
3.53.4 2026-07-24 19:02:57 bf7c7f30031888f4e796e429ab3978879485813aaca6f641c7b33e4e0945alt1 (64-b
it)
Linux kali 7.1.5+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 7.1.5-1kali1 (2026-07-29) x86_64 GNU/Linu
x
sha256sum (GNU coreutils) 9.10

kali@kali:~/SBT-DF204/Case2/original$ sha256sum History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19  History
```

A4

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT name
 FROM sqlite_master
 WHERE type='table'
 ORDER BY name;"
          name
----------------------
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits

kali@kali:~/SBT-DF204/Case2$
```

A5

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"PRAGMA table_info(urls);
 PRAGMA table_info(visits);
 PRAGMA table_info(downloads);
 PRAGMA table_info(downloads_url_chains);"
cid       name           type       notnull  dflt_value  pk
---  ---------------  ----------  -------  ----------  --
  0  id               INTEGER           0              1
  1  url              LONGVARCHAR       0              0
  2  title            LONGVARCHAR       0              0
  3  visit_count      INTEGER           1  0           0
  4  typed_count      INTEGER           1  0           0
  5  last_visit_time  INTEGER           1              0
  6  hidden           INTEGER           1  0           0
cid              name                    type     notnull  dflt_value  pk
---  ---------------------------------  -------  -------  ----------  --
  0  id                                 INTEGER        0              1
  1  url                                INTEGER        1              0
  2  visit_time                         INTEGER        1              0
  3  from_visit                         INTEGER        0              0
  4  transition                         INTEGER        1  0           0
  5  segment_id                         INTEGER        0              0
  6  visit_duration                     INTEGER        1  0           0
  7  incremented_omnibox_typed_score    BOOLEAN        1  FALSE       0
  8  opener_visit                       INTEGER        0              0
cid          name                type       notnull  dflt_value  pk
---  ----------------------  ----------  -------  ----------  --
  0  id                      INTEGER           0              1
  1  guid                    VARCHAR           1              0
  2  current_path            LONGVARCHAR       1              0
  3  target_path             LONGVARCHAR       1              0
  4  start_time              INTEGER           1              0
  5  received_bytes          INTEGER           1              0
  6  total_bytes             INTEGER           1              0
  7  state                   INTEGER           1              0
  8  danger_type             INTEGER           1              0
  9  interrupt_reason        INTEGER           1              0
 10  hash                    BLOB              1              0
 11  end_time                INTEGER           1              0
 12  opened                  INTEGER           1              0
 13  last_access_time        INTEGER           1              0
 14  transient               INTEGER           1              0
 15  referrer                VARCHAR           1              0
 16  site_url                VARCHAR           1              0
 17  tab_url                 VARCHAR           1              0
 18  tab_referrer_url        VARCHAR           1              0
 19  http_method             VARCHAR           1              0
 20  by_ext_id               VARCHAR           1              0
 21  by_ext_name             VARCHAR           1              0
 22  etag                    VARCHAR           1              0
 23  last_modified           VARCHAR           1              0
 24  mime_type               VARCHAR(255)      1              0
 25  original_mime_type      VARCHAR(255)      1              0
 26  embedder_download_data  VARCHAR           1  ..          0
cid      name         type       notnull  dflt_value  pk
---  -----------  -----------  -------  ----------  --
  0  id           INTEGER            1              1
  1  chain_index  INTEGER            1              2
  2  url          LONGVARCHAR        1              0
```

A6

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT 'urls' AS table_name, COUNT(*) AS row_count FROM urls
 UNION ALL
 SELECT 'visits', COUNT(*) FROM visits
 UNION ALL
 SELECT 'downloads', COUNT(*) FROM downloads
 UNION ALL
 SELECT 'downloads_url_chains', COUNT(*) FROM downloads_url_chains;"
     table_name       row_count
 ____________________   _________
urls                         57
visits                       74
downloads                     1
downloads_url_chains          1

kali@kali:~/SBT-DF204/Case2$
```

A7

```
kali@kali:~/SBT-DF204/Case2/original$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT id,
 datetime(last_visit_time/1000000 - 11644473600, 'unixepoch') AS last_visit_utc,
 url,
 visit_count
 FROM urls
 ORDER BY last_visit_time DESC
 LIMIT 20;"
id   last_visit_utc       url                                                
```

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT v.id AS visit_id,
        v.from_visit,
        v.transition,
        datetime(v.visit_time/1000000 - 11644473600,'unixepoch') AS visit_utc,
        u.url,
        u.title
 FROM visits v
 JOIN urls u ON u.id = v.url
 ORDER BY v.visit_time;"
```

visit_id	from_visit	transition	visit_utc	url	title
1	0	805306373	2022-04-19 13:54:18	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs...	craigslist - Google Search
2	0	805306368	2022-04-19 13:54:19	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs...	craigslist - Google Search
3	2	268435456	2022-04-19 13:54:19	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
4	3	-2147483648	2022-04-19 13:54:19	https://geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
5	4	-1610612736	2022-04-19 13:54:19	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
6	5	268435456	2022-04-19 13:54:22	https://accounts.craigslist.org/login/home	craigslist account
7	6	-1610612736	2022-04-19 13:54:22	https://accounts.	craigslist - account log

A9

```
~/SBT-DF204/Case2/working/History_working \
"SELECT v.id AS visit_id,
        v.transition,
        v.transition & 255 AS base_type,
        v.transition & 4294967040 AS qualifiers,
        u.url
 FROM visits v
 JOIN urls u ON u.id = v.url
 ORDER BY v.visit_time;"
```

visit_id	transition	base_type	qualifiers	url
1	805306373	5	805306368	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8
2	805306368	0	805306368	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8
3	268435456	0	268435456	https://www.craigslist.org/
4	-2147483648	0	2147483648	https://geo.craigslist.org/
5	-1610612736	0	2684354560	https://baltimore.craigslist.org/
6	268435456	0	268435456	https://accounts.craigslist.org/login/home
7	-1610612736	0	2684354560	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L
8	805306375	7	805306368	https://accounts.craigslist.org/login/home
9	268435456	0	268435456	https://www.craigslist.org/
10	-1610612736	0	2684354560	https://baltimore.craigslist.org/
11	268435456	0	268435456	https://post.craigslist.org/c/bal
12	-2147483648	0	2147483648	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5
13	-1610612736	0	2684354560	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type
14	805306375	7	805306368	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat

A10

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT v.id AS visit_id,
        v.from_visit,
        datetime(v.visit_time/1000000 - 11644473600,'unixepoch') AS visit_utc,
        u.url AS this_url,
        pu.url AS came_from_url
 FROM visits v
 JOIN urls u ON u.id = v.url
 LEFT JOIN visits pv ON pv.id = v.from_visit
 LEFT JOIN urls pu ON pu.id = pv.url
 ORDER BY v.visit_time;"
```

visit_id	from_visit	visit_utc	this_url	came_from_url
1	0	2022-04-19 13:54:18	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs...	
2	0	2022-04-19 13:54:19	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs...	
3	2	2022-04-19 13:54:19	https://www.craigslist.org/	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&...
4	3	2022-04-19 13:54:19	https://geo.craigslist.org/	https://www.craigslist.org/
5	4	2022-04-19 13:54:19	https://baltimore.craigslist.org/	https://geo.craigslist.org/
6	5	2022-04-19 13:54:22	https://accounts.	https://baltimore.craigslist.org/

A11

```
kali@kali:~/SBT-DF204/Case2$ sqlite3 -readonly -header -column \
~/SBT-DF204/Case2/working/History_working \
"SELECT d.id AS download_id,
        d.target_path,
        datetime(d.start_time/1000000 - 11644473600,'unixepoch') AS start_utc,
        d.total_bytes,
        d.state,
        d.tab_url,
        d.referrer,
        c.chain_index,
        c.url AS chain_url
 FROM downloads d
 LEFT JOIN downloads_url_chains c ON c.id = d.id
 ORDER BY d.start_time, c.chain_index;"
download_id target_path  start_utc  total_bytes state    tab_url    referrer  chain_index chain_ur
                                                                                          l
___________ ___________  _________  ___________ _____  _____________ ________  ___________ _______
          1 C:\Users\    2022-04-19       33844     1 https://imgur https://            0 https://
            FSCS_User\   14:56:58                    .com/dTgrkP7  imgur.com              i.imgur.
            Desktop\                                               /                      com/
            proof_of_pay                                                                  dTgrkP7.
            ment.png                                                                      png

kali@kali:~/SBT-DF204/Case2$
```

6. Timeline summary

Time (UTC)	Recorded activity	Forensic interpretation
13:54–13:58	Craigslist pages, search pages, account/login pages, and posting-related pages	The browser history records navigation through Craigslist and pages associated with an account or posting workflow. It does not independently prove that a post was successfully submitted.
From 13:58 onward	Gmail pages, including a message titled “cheaper than Rx supplements” and compose-related pages	The records show visits to Gmail pages and a message-related page. They do not establish whether an email was sent or who authored it.
14:56:58–14:57:10	Download record for proof_of_payment.png, associated with Imgur	The database records a download target path, byte count of 33,844, and an associated URL chain pointing to https://i.imgur.com/dTgrkP7.png. The file's contents were not established from the browser metadata alone.
14:58:59–14:59:21	Google search for “blockchain explorer,” followed by Blockchain.com explorer and	The history records navigation to cryptocurrency-explorer pages. This alone does not establish

	cryptocurrency transaction/address pages	ownership of an address or completion of a transaction.
14:59:38–15:00:21	Google search for “kraken,” Kraken sign-in, and a device-approval page	The records show visits to Kraken-related pages, including sign-in and device-approval pages. They do not independently confirm a successful login.
15:00:21–15:02:05	Kraken trading, instant, funding, and ledger pages; a redirect to a transaction page on mempool.space	The browser records navigation through these pages. It does not by itself prove that a trade, deposit, withdrawal, or cryptocurrency transfer was completed.
15:02:32–15:02:56	Further visits to the Gmail message-related page	The history records additional navigation to the page near the end of the observed timeline.

7. Attribution Assessment

Item	**Assessment**
Confidence level (low / moderate / high)	Moderate for identifying the browsing activity recorded in the Chrome History database; low for attributing the activity to a specific person.
Basis for confidence	The analysis used a preserved original file and a matching-hash working copy. Four SQL queries examined the browsing timeline, transition values, recorded navigation relationships, and download URL chain. The results showed timestamps, URLs, page titles, and a download record that could be cross-checked against one another.
Limitation 1	The History database does not independently identify the person operating the device or prove that the account holder personally performed each recorded action.
Limitation 2	Browser history records page navigation but does not, by itself, prove that a login succeeded, an email was sent, a Craigslist post was submitted, or a cryptocurrency transaction was completed.
Limitation 3	The analysis was based primarily on browser-history metadata. The contents of

	proof_of_payment.png, relevant email contents, server logs, and independent transaction records were not established by this database analysis alone.
Alternative explanation(s)	The activity could have been performed by the device owner, another person using the device, someone with access to the browser profile, or an automated process. Some navigation could also have resulted from redirects, embedded content, or links opened by another page. These are possibilities, not established facts.
Shared device / browser profile	The available database does not establish whether the device or browser profile was shared. Other users may have accessed the same profile. Further evidence, such as operating-system login records and profile information, would be needed to assess this possibility.
Redirects, cached or missing content	Redirects may explain some navigation relationships, including the recorded route to a cryptocurrency transaction page. The History database may preserve URLs and titles even when page content is no longer available. Missing content limits what can be concluded about the exact information displayed or actions performed.

Attribution conclusion

The findings provide moderate confidence that the analyzed browser profile recorded the listed browsing activity, but they provide low confidence in attributing that activity to a particular individual without corroborating evidence. The assessment distinguishes recorded browser events from conclusions about identity, intent, and completed actions. Further examination of the downloaded image, relevant email artifacts, operating-system records, and independent transaction evidence would help strengthen or weaken the attribution hypothesis.